Answer Key & Teacher Notes

Session 1 — Finding the Median · Session 2 — Median of an Even Data Set

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Finding the Median

TALK PROMPTS

- Why must you put the values in order before you can point at the median?
- How does ordering the list first make the middle value easy to point at?

PARTNER TASK

Pizza Slices. One partner orders the seven values; the other crosses off one value from each end until one is left. Switch and repeat with Trail Lengths, then test what happens when the greatest value grows.

WATCH FOR

- Taking the middle value of the list as written instead of the ordered list.
- Choosing the largest value or the most common value as the median.
- Thinking a very large new greatest value pulls the median up.

SESSION 2 · TEACHER NOTES

Median of an Even Data Set

TALK PROMPTS

- Why does an even number of values need two middle numbers instead of one?
- Why do you add the two middle values and then divide that sum by 2?

PARTNER TASK

Shoe Sale. Partners order 28, 42, 36, 24, 48, 32 together, then one finds the two middle values and the other averages them (34). Check: three values sit below 34 and three above.

WATCH FOR

- Giving two numbers as the median instead of averaging them.
- Averaging the two middle values of the list as written instead of the ordered list.
- Thinking the median must be one of the data values.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.SP.B.5c	1	1	2	1	2	3	2	3	2	1	1
S1 · Form B	6.SP.B.5c	1	1	2	1	2	3	2	3	2	1	1
S2 · Form A	6.SP.B.5c	1	1	2	1	2	3	3	2	2	1	1
S2 · Form B	6.SP.B.5c	1	1	2	1	2	3	3	2	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

median: The **middle** value of a data set in order.

data set: A **group** of values you collect.

order: To list values from **least** to greatest.

- Always **order** the values before you find the median.
- With an odd number of values, the median is the **one** value in the middle.
- Changing the greatest value does not **change** the median.

Example: Order the values: 2, 4, **6**, 7, 9. Cross off one value from each end until **one** is left. Answer: Median = **6**

YOU TRY

1. a. 5 b. 10

2. Order: 11, 11, 12, 12, 13, 14, 15. Median = 12 years

EXIT TICKET

1. a. 6 b. 16

2. Order: 1, 2, 2, 3, 3, 3, 5. Median = 3 (the 4th value)

3. **B** (order the values)

PRACTICE FORM A

1. Order: 9, 11, 13, 14, 17, 19, 20, 22, 25. Median = 17 (the 5th value)

2. a. 30 (order 15, 20, 30, 35, 45) b. 2

3. a. 7 (order 2, 4, 6, 7, 9, 11, 12) b. 7 km, no change

4. Order: 29, 34, 38, 42, 45, 47, 51. Median = 42 slices

5. No. Order first: 12, 18, 23, 27, 31. The median is 23, not 27.

6. Order: 14, 26, 32, 38, ?. The median is the 3rd value, 32 cm. The unknown sits at the right end, so it never moves into the middle.

7. Both medians are 29. Only the greatest value changed, and the middle value stays in the 3rd position.

8. a. Order: 55, 57, 58, 59, 61, 62, 64; median 59 in. b. 59 in. (no change) c. The median ignores how far the tallest value is; it only uses the middle position.

9. The median is the middle of the ordered list, not the largest value. Order: 2, 4, 6, 7, 9. Median = 6.

10. **B** (6)

11. **C** (15)

Self-check word **ORDERING**

PRACTICE FORM B

1. Order: 8, 10, 12, 15, 16, 18, 21, 23, 27. Median = 16 (the 5th value)

2. a. 35 (order 10, 25, 35, 40, 50) b. 2

3. a. 8 (order 3, 5, 7, 8, 10, 13, 15) b. 8 km, no change

4. Order: 31, 36, 40, 44, 47, 49, 53. Median = 44 bagels

5. No. Order first: 16, 22, 28, 35, 41. The median is 28, not 35.

6. Order: 18, 27, 33, 45, ?. The median is the 3rd value, 33 cm. The unknown sits at the right end, so it never moves into the middle.

7. Both medians are 22. Only the greatest value changed, and the middle value stays in the 3rd position.

8. a. Order: 57, 59, 60, 61, 63, 64, 66; median 61 in. b. 61 in. (no change) c. The median ignores how far the tallest value is; it only uses the middle position.

9. The median is the middle of the ordered list, not the largest value. Order: 3, 5, 7, 8, 10. Median = 7.

10. **C** (7)

11. **B** (5th value)

Self-check word **POSITION**

GUIDED NOTES

even number of values: A data set with **two** values in the middle.

middle values: The two values left after you cross off from both **ends**.

average of two numbers: Add the two numbers, then **divide** by 2.

- With an even number of values, the median is **halfway** between the two middle values.
- The median does not have to be one of the **values** in the set.

Example: Order: 8, **12**, **15**, 19. Add the middle values and divide by 2: 27 ÷ 2. Answer: Median = **13.5**

YOU TRY

1. a. 6 b. 15

2. Order: 5, 7, 8, 10, 12, 14. (8 + 10) ÷ 2 = 9 pounds

EXIT TICKET

1. a. 9 b. 5.5

2. Order: 1, 2, 2, 3, 4, 5. (2 + 3) ÷ 2 = 2.5

3. **A** (28 and 32)

PRACTICE FORM A

1. Order: 78, 82, 84, 86, 88, 91, 94, 96. Middle values 86 and 88; median = (86 + 88) ÷ 2 = 87 s

2. a. 20 (order 12, 15, 18, 22, 25, 29; (18 + 22) ÷ 2) b. No

3. Order: $0.95, $1.25, $1.45, $1.75. ($1.25 + $1.45) ÷ 2 = $1.35; median = 1.35 dollars

4. Order: 24, 28, 32, 36, 42, 48. (32 + 36) ÷ 2 = 34 dollars

5. a. 45 b. 40 (order 20, 25, 35, 40, 45, 50, 65) c. down

6. No. Now there are 6 values, so the median is the average of the two middle values: (18 + 22) ÷ 2 = 20.

7. a. 6 mm (order 2, 4, 5, 7, 9, 11) b. 7 mm c. The new value only shifts the middle position by half a value; how big it is does not matter.

8. a. 31.5 min (order 22, 26, 28, 30, 33, 35, 37, 41; (30 + 33) ÷ 2) b. With an even count, the median is halfway between the two middle values, so it can be a number that is not in the list.

9. A median is one number. Renata skipped averaging the two middle values: (12 + 15) ÷ 2 = 13.5.

10. **C** (8)

11. **B** (average of the two middle values)

Self-check word **AVERAGED**

PRACTICE FORM B

1. Order: 69, 72, 74, 77, 81, 85, 88, 90. Middle values 77 and 81; median = (77 + 81) ÷ 2 = 79 s

2. a. 21 (order 11, 13, 16, 26, 27, 31; (16 + 26) ÷ 2) b. No

3. Order: $1.85, $2.15, $2.35, $2.65. ($2.15 + $2.35) ÷ 2 = $2.25; median = 2.25 dollars

4. Order: 22, 26, 30, 42, 44, 52. (30 + 42) ÷ 2 = 36 dollars

5. a. 40 b. 35 (order 15, 20, 30, 35, 40, 55, 60) c. down

6. No. Now there are 5 values, so the median is the 3rd value: 30.

7. a. 7 mm (order 1, 3, 6, 8, 10, 12) b. 8 mm c. The new value only shifts the middle position by half a value; how big it is does not matter.

8. a. 33 min (order 24, 27, 29, 32, 34, 37, 39, 43; (32 + 34) ÷ 2) b. With an even count, the median is halfway between the two middle values, so it can be a number that is not in the list.

9. A median is one number. Malik skipped averaging the two middle values: (9 + 14) ÷ 2 = 11.5.

10. **B** (19)

11. **C** (5th and 6th values)

Self-check word **BALANCED**

SHEET 1 · FINDING THE MEDIAN

1. a. 6 b. 12 c. 30 d. 5

2. a. 9 students b. Ordered: 6, 6, 7, 7, 8, 8, 8, 9, 10; the 5th value is 8 hours.

3. Ordered: 4, 6, 7, 9, 11, 12, 15. Median = 9 miles.

4. 85 points. Ordered, the four known scores are 75, 82, 88, 90. The median is the 3rd of 5 values, and 85 is not one of them, so the fifth score must be 85 (75, 82, 85, 88, 90).

5. Any 5 different values with 12 in the middle when ordered, e.g. 3, 8, 12, 15, 20 (two values less than 12, two greater).

6. No. Ordered 4, 6, 9, 11, 100 still has 9 in the middle. Changing the greatest value does not move the middle value, so the median stays 9.

SHEET 2 · MEDIAN OF AN EVEN DATA SET

1. a. 6 b. 16 c. 8.5 d. 37.5 e. 15 f. 6.5

2. a. 1 and 2 (the 5th and 6th values) b. (1 + 2) ÷ 2 = 1.5 cousins

3. Ordered: 5, 7, 8, 10, 12, 15. (8 + 10) ÷ 2 = 9 pounds.

4. 14. The two middle values are 10 and the missing value. Their sum is 12 × 2 = 24, so 24 − 10 = 14 (order 6, 10, 14, 20 checks).

5. She did not order the data first; 2 and 10 are the middle values of the list as written. Ordered: 2, 6, 8, 10, so the median is (6 + 8) ÷ 2 = 7.

6. **A** (13.5)

7. **A** (28 and 32)

October 2026 conceptual additions — teacher key

Session 1 · Finding the Median

Ordered data: 3, 3, 7, 8, 9. Median: 7. Both 3s are separate observations; removing one changes the data and the middle positions.

Extension: No; the ordered values become 3, 3, 7, 8, 90. The third value remains 7.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Median of an Even Data Set

Ordered data: 3, 5, 8, 12. Median = (5 + 8) ÷ 2 = 6.5. The median need not be an observed value.

Extension: Example: 2, 6, 7, 10; the middle pair averages to 6.5. Other valid sets are accepted.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.